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From Literature to Knowledge Graphs: Automated Extraction and Retrieval of Life Cycle Assessment Data with Large Language Models

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Abstract

The large and steadily growing volume of scientific literature poses challenges for accessing and utilizing data due to its unstructured nature. Life cycle assessment (LCA), in particular, relies on high-quality life cycle inventory (LCI) data to quantify the environmental impacts of products or services across their life cycle, yet manual extraction of such data is time-consuming and labor-intensive. This study presents a framework that integrates large language models (LLMs) with knowledge graph (KG) to extract and manage LCA data from published studies. A retrieval-augmented generation (RAG) pipeline is developed to mine three core data types from full-text articles: LCI data, life cycle impact assessment (LCIA) results, and LCA modeling assumptions. The extracted information is normalized and mapped to an ontology-driven, LCA-oriented knowledge graph (LCAKG) implemented in Neo4j. To support user interaction, an LLM-based question-answering system translates natural language queries into executable graph queries, allowing users to retrieve rich information without prior knowledge of KG schemas. The framework is evaluated using a case study of LCA studies on chemical production. The results demonstrate high semantic accuracy in data extraction, with F1-scores ranging from 73.54% to 93.34%. Query performance is significantly improved by combining similarity search with text-to-cypher reasoning, increasing the F1-score from 56.98% (baseline) to 75.18%. The proposed framework enhances the accessibility and interoperability of LCA domain data and provides a scalable foundation for large-scale knowledge synthesis to support sustainability research.

Key words

Life cycle assessment; Large language models; Machine Learning; Data Extraction; Knowledge graph

1. Introduction

With growing awareness of climate change and its effects, the use of life cycle assessment (LCA) is steadily increasing in research and industry to inform decarbonization policies and sustainable decision-making (Gaurav et al., 2021). Performing an LCA is often a data-driven process (Mittal et al., 2018), requiring extensive data for both life cycle inventory (LCI) development and life cycle impact assessment (LCIA) modeling. An ever-growing volume of LCA data released in scientific literature and industrial reports, including impact results and detailed material and energy data, can serve as reliable sources for environmental benchmarking (Schlegl et al., 2019) and for filling inventory data gaps when primary data is unavailable (Saavedra-Rubio et al., 2022a; Spreafico & Russo, 2020). However, the availability of data does not constitute an easily accessible and readily usable database. By contrast, efficient extracting and retrieving LCA data from literature remains a significant challenge due to its unstructured natural language text formats (e.g., PDFs) and the heterogeneous data storage formats (Kuczenski et al., 2016). Manual data curation is time-consuming and infeasible for large-scale practice (Prioux et al., 2021). Therefore, developing an automated pipeline for LCA data mining is essential for constructing a structured and accessible database that can maximize the value of underutilized data in existing LCA studies.

Traditionally, natural language processing (NLP) techniques have been employed to extract LCA-related data from text-based documents through a series of operations, such as text processing, entity recognition, and text classification (Gachkar et al., 2025; Hu et al., 2025). However, these approaches are typically task-specific, labor-intensive, and require substantial expertise in programming and computer science (Zheng et al., 2023). Recent advancements in generative artificial intelligence (AI) algorithms have brought large language models (LLMs) to the forefront in handling NLP tasks, offering potential solutions to enhance the efficiency and reliability of unstructured data mining (Y. Ye et al., 2024). LLMs, such as GPT-4, are deep learning models trained on large text corpora through pre-training and fine-tuning techniques (Turhan, 2023). These models can overcome some of the limitations of traditional NLP approaches due to their high accessibility (e.g., via web

browsers or APIs) and ease of use (requiring no additional training by users). Acknowledging these advantages, researchers have increasingly applied the LLMs to extract domain-specific knowledge and structured data from documents to accelerate research and innovations in areas such as chemistry (Schilling-Wilhelmi et al., 2025; W. Zhang et al., 2024; Zheng et al., 2023), medicine (Cao et al., 2024; Dagli et al., 2024), and material science (Dreger et al., 2025; Gupta et al., 2024a; Y. Ye et al., 2024).

Similar applications have also emerged in the LCA domain (Chen et al., 2024; Goridkov et al., 2024; Kumar et al., 2025; Oladeji & Mousavi, 2023; Turhan, 2023; Zhao et al., 2025). For example, Goridkov et al. (2024) developed an LLM-based framework to retrieve sustainability-related insights from LCA reports, including scope of analysis, environmental hotspots, and design strategies. The findings suggest that the LLMs can effectively interpret and summarize complex LCA documents, thereby enhancing their accessibility for designers and other non-experts. However, LLMs are prone to generating responses that deviate from factual accuracy, often termed as hallucination, which pose significant challenges to user trust and the reliability of AI systems. To mitigate the possibility of hallucination, Kumar et al. (2025) and Zhao et al. (2025) fine-tuned open-source LLMs as domain-specific question-answering models to extract LCA-related data from unstructured texts. One major limitation of fine-tuning, however, is the requirement for large amount of annotated data, and the data curation process can take several months for researchers to complete (Jami et al., 2024; Kumar et al., 2025). Alternatively, prompt engineering can minimize hallucination by formulating prompts for LLMs through techniques such as zero-shot and few-shot learning. Studies have demonstrated that even using general-purpose LLMs with prompt engineering can achieve high performance in data mining tasks (Luo et al., 2025; Polak & Morgan, 2024; Zheng et al., 2023), thus considerably reducing deployment time and computational cost compared with fine-tuning. Nevertheless, prompt engineering remains underutilized in the field of LCA, with only a few studies emphasizing its importance and potential applications in extracting LCA domain data.

Retrieval augmented generation (RAG) is a state-of-the-art technique that integrates the LLMs with external knowledge sources (typically stored in a vector database) to generate more accurate and grounded responses. Instead of processing the full content, RAG retrieves semantically relevant chunks from the knowledge base as context for querying the language model, making it a fast and cost-effective strategy for data mining from large text corpora (Schilling-Wilhelmi et al., 2025). A knowledge graph (KG) is a structured representation of entities and their relationships of a specific domain in the format of nodes and edges. Recently, KGs have gained significant attention for managing vast-volume and complex LCA data due to their abilities to handle semantic relationships, offer scalability, and support intelligent querying (Greif et al., 2025; Peng et al., 2024; Tu et al., 2024; Y. Ye et al., 2024). Despite these advantages, to the best of our knowledge, no existing studies have integrated RAG-based data extraction directly with structured, machine-readable KG representations for LCA data. Herein, we present a holistic framework that leverages the GPT-based LLM and KG to extract and manage LCA data from scientific literature. We first develop a RAG-based pipeline to automate the extraction of three types of information from LCA literature, namely, LCI inventory data, LCIA results, and LCA modeling assumptions (e.g., system boundaries, functional units, impact assessment methods, and case study geography). The extracted data are then mapped to an LCA-oriented knowledge graph (LCAKG) through the pre-defined ontology to facilitate efficient querying and reasoning. To evaluate the practicality and feasibility of the proposed framework, we conduct a case study using 107 LCA papers focused on chemical synthesis.

Our contributions are highlighted in four key areas: (1) We propose a framework that integrates LLMs and KG to achieve efficient LCA data extraction and management from LCA publications. Through this framework, we can easily convert unstructured LCA knowledge from PDF documents into structured semantic data stored in the graph database. This framework can be extended and customized to extract other types of knowledge based on specific requirements; (2) To evaluate the performance of the proposed pipeline, we build two LCA domain ground-truth datasets: one consisting of meticulously labeled entities extracted from the literature, and the other comprising question-answer pairs derived from

the graph dataset. These datasets fill critical data gaps in the LCA domain by providing a valuable benchmark for comparing and validating other AI/ML applications; (3) We apply the proposed framework to process 107 LCA papers on chemical production, extracting a total of 146 inventory tables and LCA outcomes across various chemical synthesis pathways. The extracted database can serve as a secondary data source for future LCA studies; and (4) We improve retrieval quality and efficiency by combining LLMs with similarity search in the query system, allowing users to ask natural language questions to retrieve information more effectively and accurately. The remainder of this paper is structured as follows: Sector 2 outlines the methods and data used to develop and evaluate the framework. Sector 3 presents the results and discussion of the case study, with a focus on pipeline performance, current challenges, and limitations. Finally, Sector 4 summarizes and concludes the study.

2. Methods

2.1 LCAKG development framework

The proposed framework follows a multi-stage process, as illustrated in Figure 1. First, we developed an RAG-based pipeline to extract LCA domain data (e.g., LCI data, LCIA results, and LCA modeling assumptions) from the full text of journal articles. Next, we manually consolidated the LLM outputs into a tabular dataset, which was then mapped into a LCAKG using the predefined ontology. Finally, we built an intelligent retrieval system based on the LCAKG that enables users to efficiently query data through natural language question-answering. The detailed processes are described below.

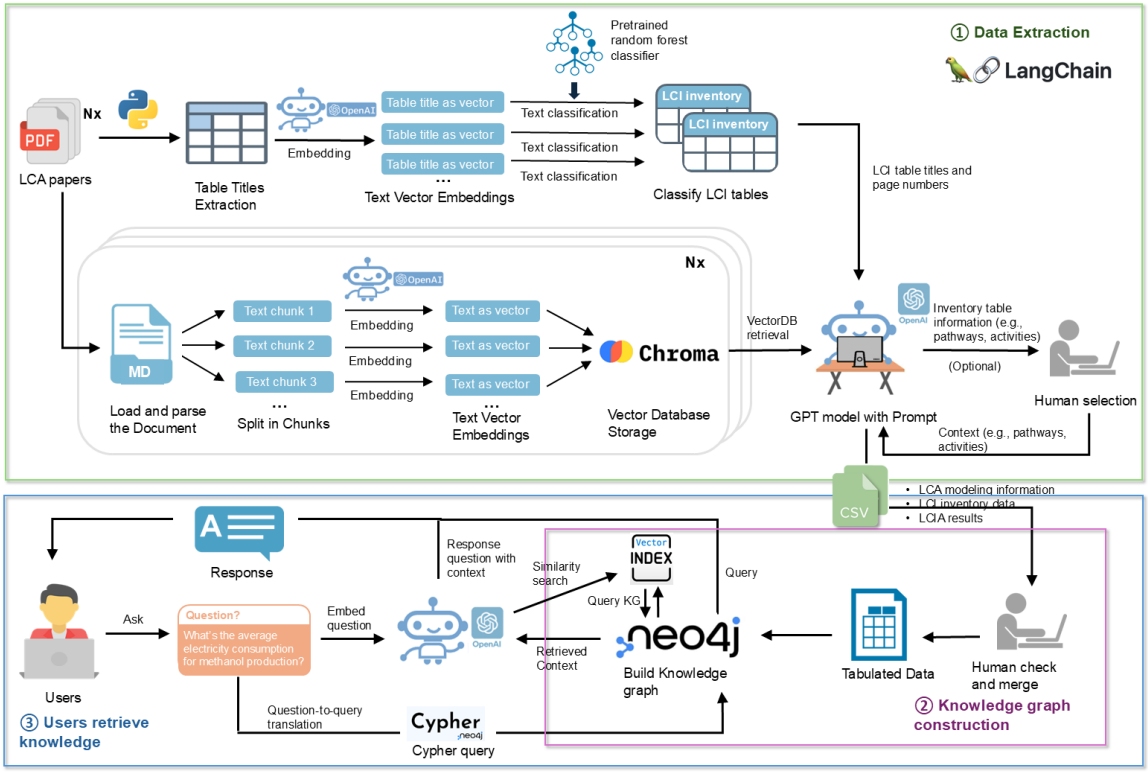


Figure 1. Schematics of the LCAKG development framework.

Process 1: Domain data extraction

The idea behind the data extraction pipeline is that users only need to provide LCA documents in PDF format, and the pipeline will automatically convert all relevant data from text into structured outputs. In this study, we handled the extraction of LCI inventory data separately from that of LCIA results and LCA modeling assumptions due to their heterogeneous structural complexity and data formats. Typically, LCIA results and LCA modeling assumptions (e.g., system boundaries, functional units, and impact assessment methods) are presented in flat tables or textual contexts with standardized keywords and consistent phrasing, making them relatively easier for language models to retrieve. In contrast, LCI inventory data is often stored in more complex data structures (e.g., nested tables or flat tables with multi-sections), where inputs and outputs from single or multiple unit processes are presented in an aggregated format. Furthermore, different papers describe their inventory

tables with varying titles, such as “LCI data”, “Life cycle inventory”, “Input and output data”, “Bill of materials”, and “Resource consumption and emissions”, which makes it more difficult for the LLMs to accurately identify and extract the relevant data.

To address these challenges in LCI data extraction, we first conducted binary text classification with embeddings to distinguish inventory table titles from other types of tables. Embedding is a well-established technique in NLP that maps text into high-dimensional vectors, enabling the quantification of both semantic and syntactic information (da Costa et al., 2023). First, we extracted all table titles from each document by regular expressions and then transformed them into vector representations using the OpenAI embedding model (text-embedding-3-small), as it demonstrates superior performance in classification tasks (Korade et al., 2024a). Next, we utilized a pre-trained Random Forest Classifier (RFC) to predict whether each embedded vector represents an inventory table title or not. This classifier was trained on a balanced dataset of 156 table titles extracted from 50 randomly selected peer-reviewed LCA studies on Google Scholar, as detailed in the supplementary materials (SI). Each title was manually labeled as True or False based on whether it referred to an LCI inventory table. The titles were then embedded, and the dataset was randomly split into 80% for training and 20% for testing. By applying this classifier, we identified the locations of LCI inventory tables by their table titles and page numbers, which were then used as part of the query to guide accurate LLM retrieval.

To enable the language model to efficiently process each paper, we developed an RAG system based on the LangChain framework (https://www.langchain.com/), following three main steps: indexing, retrieval, and augmentation and generation. Indexing is to transform input data into embeddings and store them in a vector database for retrieval. In this study, we first converted the PDF version of LCA documents into the Markdown format using the PyMuPDF4LLM Python package, as Markdown can preserve the documents’ original layout and semantic structure (Moreira-Filho et al., 2025). The ingested content was cleaned by removing references and acknowledgements, which are unlikely to contain LCA data, and then divided into chunks using recursive character text splitting based on sentence separators.

We used the same OpenAI embedding model to generate embeddings for each chunk and stored them in Chroma DB, an open-source vector database. To improve retrieval accuracy, we created a separate vector database for each paper, with metadata specifying the document ID and the page number associated with each chunk.

In the retrieval step, the vector store is linked to a retriever to fetch the most relevant chunks by computing cosine similarity between the user query embeddings and the stored chunk embeddings. These retrieved chunks are then used to augment the original query and prompts, enabling the LLM to generate context-aware responses. These steps can be achieved using LangChain's chains, which serve as the core components connecting the LLM, retriever, and prompts (Topsakal & Akinci, 2023). Given the data heterogeneity mentioned above, we employed two chains from the LangChain library, namely Conversational Retrieval Chain and RetrievalQA. In this study, retrieving LCI data from inventory tables was designed as a two-step process. First, the LLM interpreted table metadata (e.g. table titles, headers, and stub labels), which often contained contextual information such as unit processes, technological pathways, or scenarios presented in the table. This context was then used as intermediate structural cues to guide the extraction of inventory data. The Conversational Retrieval Chain was chosen as the preferred choice, as it incorporates historical context to enhance conversational interactions (Prem Jacob et al., 2024). In cases where the LLM fails to accurately interpret table metadata, we also provided a human-in-the-loop option that allows users to interactively guide the LLM in extracting LCI data through this chain. In contrast, LCIA results and LCA modeling assumptions were extracted directly from the retrieved documents by prompts. RetrievalQA was selected for its ability to generate answers based on the retrieved content and provide source references (B & Purwar, 2024).

We used OpenAI's GPT-4o model for text data extraction. This cloud-based model offers flexible API access and demonstrates exceptional performance in handling text-based tasks, with fast processing and cost efficiency (Islam & Moushi, 2024). Prompts are the input text given to the LLM, which can be crafted through prompt engineering following three fundamental principles: minimizing hallucinations, providing detailed instructions, and

requesting structured output (Zheng et al., 2023). Following these rules, we utilized a few-shot prompting (i.e., providing examples to enable in-context learning) to guide the language model in LCA data extraction, as illustrated in the SI. The LLM was instructed to produce structured outputs in JSON format, which were then converted into CSV files for further processing. The LLM temperature was set to 0 to ensure deterministic results, with other hyperparameters summarized in Table S3. To improve the generality and reusability of the data extraction pipeline, we designed the prompts to be applicable across various LCA systems. Besides, we developed a graphical interface using the open-source library Streamlit to enable users to interact with the tool, as shown in Figure S3.

Process 2: Knowledge graph construction

The main purpose of developing the LCA-oriented knowledge graph (LCAKG) is to represent relationships within LCA domain data and bridge the gap between heterogeneous data formats across diverse LCA studies. As mentioned above, we extracted three types of data from the literature, including LCI inventory, LCIA results, and LCA modeling assumptions, with detailed contents and descriptions for each type summarized in Table 1. We conducted human validation to ensure the correctness and consistency of the LLM outputs. For LCI inventory data, we compared the retrieved results with the original sources to fill in missing data and correct inaccuracies. For LCIA results and LCA modeling assumptions, we primarily focused on resolving missing information (i.e., instances labeled as 'Not mentioned' by the LLM) through cross-checking with the source documents. All revised structured data were then merged into a tabulated dataset. Given the inherent variability observed in the literature (e.g., differences in system boundary descriptions), we performed data normalization following the rules outlined in the SI to enhance dataset interoperability.

Table 1. Main types, contents, and descriptions in the LCA-oriented knowledge graph.

Type	Content	Description
LCI Inventory	Activity	A unit process represents the smallest element in the product system for which input and output data are

		quantified.
	Flow	An input or output substance to/from an activity.
	Reference Product	The main output of a unit process, for which a change in demand directly affects the required inputs and production volumes of by-products/wastes of that process.
LCIA Results	LCIA Method	A method used to quantify the environmental impacts of a product system.
	Impact Category	A class of environmental issues of concern to which impact results may be aligned.
	Impact Results	A quantified result representing the magnitude of impact within a specific impact category.
LCA Modeling Assumptions	System Boundary	A set of criteria defining which unit processes are included in the product system being assessed.
	Functional Unit	A quantified reference that defines the function of the product system and provides a basis for comparison.
	Geography	The physical or regional context (e.g., country, region, or specific site) in which the LCA study is conducted or modeled.
Others	Sector	A broad category of activities that share common characteristics, defined under the ISIC classification system.

During the construction of the knowledge graph, the ontology serves as a structural framework that defines the concepts, entities, and their relationships within a specific domain (Peng et al., 2024). To ensure the generality and reusability of the proposed ontology, we drew inspiration from existing graph models that provide a consensus-based semantic description of core elements in LCA domain knowledge (Ghose et al., 2022; Janowicz et al., n.d.; Kuczenski et al., 2016), as illustrated in Figure 2. Using the normalized LCA dataset and the established ontology, we developed LCAKG in Neo4j, a state-of-the-art graph database management system (GDBMS). We employed Neo4j Desktop to create and manage the project, which support execution both locally and remotely via Neo4j servers. The process primarily involved creating nodes, assigning their properties, and establishing relationships

through the Cypher query language, with the Neo4j Python driver used to interface with the GDBMS.

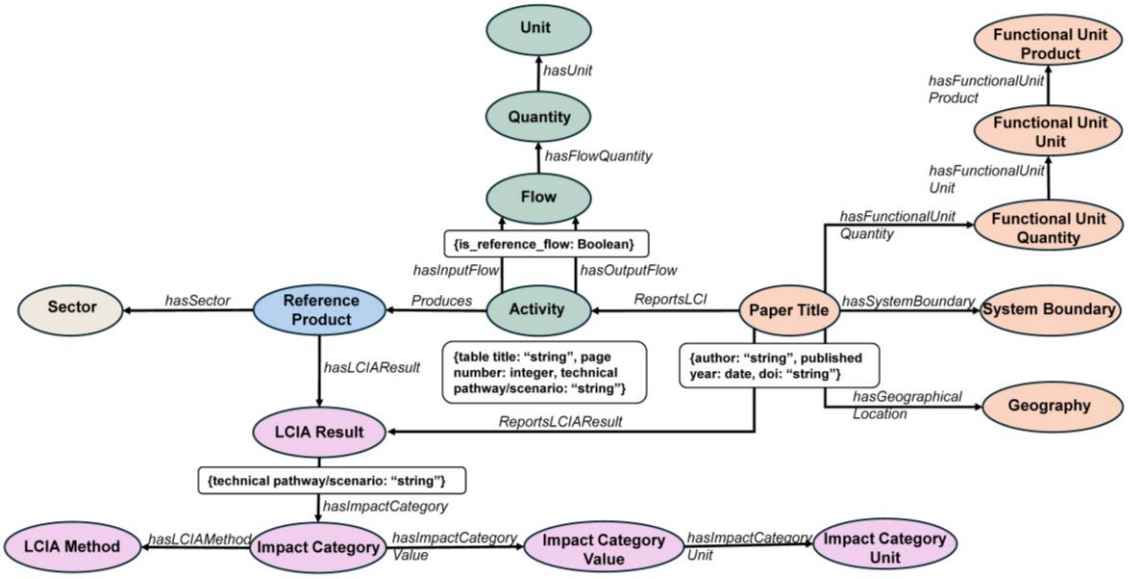


Figure 2. The ontology model of LCA-oriented knowledge graph.

Process 3: Users retrieve knowledge

The most direct method to traverse node-edge relationships in the Neo4j graph database is through Cypher queries. However, this approach requires expertise in domain-specific schemas and familiarity with graph-query constructs, which can pose a steep learning curve for non-expert users (Yang et al., 2026). To simplify user interaction with LCAKG, we integrated LLMs into our system, allowing users to retrieve information by asking natural language questions, such as "What are the global warming potential (GWP) results of a product?" or "What are the inputs and outputs involved in manufacturing a product?". This approach follows an RAG paradigm; instead of relying on a conventional vector database, it leverages a graph database as the external knowledge base to enhance the output of LLMs, commonly referred to as GraphRAG. Here, we developed a graph-based question answering system (GraphQA) that utilizes LangChain's GraphCypherQAChain and a Neo4j schema description to translate natural language queries into Cypher queries. A main challenge is that

accurate Cypher generation requires precise alignment with node labels, properties, and instance values, where even minor variations in user queries (e.g., casing or punctuation) can lead to execution failures and semantic mismatching (Yang et al., 2026).

To address this limitation, we further implemented a graph-based multi-agent system (GraphMAQA) that integrates similarity search with GraphCypherQAChain to enhance the LLM's ability to generate accurate Cypher queries. The workflow proceeds as follows: when a user query is received, the orchestrating agent (router) first determines whether it aligns with the database schema or require semantic alignment via similarity search. If the query is determined to contain exact schema terms (e.g., node labels, relationship types, or property names defined in the schema), it is routed directly to the Cypher agent, where GraphCypherQAChain connects the LLM with the Neo4j schema to generate and execute Cypher queries. If not, the query is embedded to retrieve semantically similar elements through similarity search on vector indexes, encoded from node labels and their properties using the OpenAI embedding model (text-embedding-3-small). These retrieved elements are then used to rewrite the user query before passing it to the Cypher agent to produce results. The entire workflow is illustrated in Figure S4, and the agent-based system was implemented using LangGraph (https://www.langchain.com/langgraph), with the GPT-4o model performing the text-to-Cypher tasks.

2.2 Case study

To demonstrate the applicability and functionality of the proposed framework, we conducted a case study on LCA studies of chemical production. We selected 24 target chemicals, with the full list shown in SI. The chemical names were combined with keywords such as "life cycle assessment," "life cycle analysis," "LCA," and "environmental assessment" to search the scientific literature published between 2010 and 2024 in Web of Science, Scopus, Science Direct, Springer, and Google Scholar. We manually screened the title, keyword, and abstract to remove duplicates and irrelevant studies. As a result, 107 research papers were selected in this study (see SI), and their full texts were procured in PDF format for pipeline processing. To simplify the data curation step, we focused solely on extracting the global warming

potential (GWP) results and ignored other impact categories. In addition, we only extracted gate-to-gate LCI data of the chemical production process (e.g., methanol synthesis), excluding upstream stages such as raw material production and transportation when these were reported separately in the literature.

2.3 Pipeline evaluation

The ground truth is a labeled dataset that serves as a necessary benchmark for evaluating algorithm performance (Fang et al., 2012). In this study, we built two ground truth datasets to determine the performance of data extraction in Process 1 and knowledge retrieval and question-answering (QA) in Process 3. For Process 1, the performance of the RFC in classifying LCI table titles was first evaluated and compared with baseline approaches (i.e., LLM-based and rule-based approaches), as detailed in the SI. The proposed pipeline was then applied to a dataset comprising 65 full-text LCA studies on bio-based chemicals, as identified in our previous literature review (Tang et al., 2025). The resulting structured data were meticulously cross-checked against the original sources to establish the ground truth. To minimize bias, we developed detailed rules (shown in SI) to guide human annotators in labeling entries from the papers. For example, we marked the ground truth as "Not mentioned" if the information was neither explicitly stated nor reasonably inferable from the text. For Process 3, we utilized the entire results from the case study to construct a dataset of 50 question-answer pairs. Each Q&A pair consists of a randomly sampled multi-hop natural language question derived from the graph schema, along with a manually annotated ground-truth answer based on the graph database.

BERTScore is an automatic text-generation evaluation metric that uses contextual embeddings from a pretrained transformer encoder (often BERT/RoBERTa) to align tokens between a candidate and reference, and scores their semantic similarity via cosine similarity (reported as precision/recall/F1) (Zhang et al., 2020). In this study, BERTScore was applied to evaluate the performance of Process 1 and Process 3. We calculated BERTScore between the model's output and the reference answer, returning precision, recall, and F1 score that quantify how well the generated text captures the semantic meaning of the reference. A

special rule was applied in Process 1 when multiple instances were presented within the same content type (e.g., multiple GWP scenarios in one paper), in which case they were aggregated as a single entity for evaluation. Besides, the extracted LCI table was assessed at the row level to calculate resulting metrics, which were then averaged across rows to measure the overall performance. However, a key limitation of this metric is that it does not account for numerical fidelity (Kumar et al., 2025), meaning it cannot accurately capture discrepancies between estimated values and their ground truth (e.g., 1 kg CO2-eq versus 0.01 kg CO2-eq still yield a high similarity score). To address this issue, we additionally calculated mean absolute percentage error (MAPE) of the numerical components in the LCIA results, and LCI data using the equation below, where y_i is the ground truth i^{th} value, $\hat{y}_i$ is the LLM generated i^{th} values, and n denotes the total number of observations.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100$$

3. Results and Discussion

3.1 Evaluation of LCA data mining performance

We began our performance analysis by first evaluating the results of LCI data extraction. The initial step in this process was to recognize the relevant data regions containing extractable LCI information based on their table titles. This pre-filtering step is crucial when extracting data from lengthy documents (e.g., research papers) (Gupta et al., 2024; Hu et al., 2025; Zheng et al., 2023), as it avoids unnecessary prompting of the LLM and ensures that data are extracted correctly from the relevant texts. Our results showed that the RFC outperformed both the LLM prompting-based and rule-based classifiers, achieving the highest precision, recall, and F1 scores of 0.97 (Figure 3a). While simple keyword matching often proves insufficient (Zheng et al., 2023), the relatively lower performance of the LLM-based model can be attributed to two main factors: (1) the classification schema (LCI/non-LCI) is highly domain specific, and the general-purpose LLM lacks sufficient domain knowledge and is constrained by idiosyncratic task-design choices (e.g., few-shot prompting with only three

examples per class), leading to errors such as mislabeling 'Bill of materials of the bus model' as non-LCI; and (2) unlike tasks involving rich textual context for effective reasoning (Sushil et al., 2024; Zheng et al., 2023), the LCI table titles are short sentences and provide limited semantic cues for LLMs to accurately infer labels. In contrast, the RFC learns decision boundaries in an embedding space that encodes the semantic meaning of LCI table titles. This ML-based approach also effectively resolves the overfitting issue through a bagging mechanism (Korade et al., 2024), resulting in overall superior classification performance.

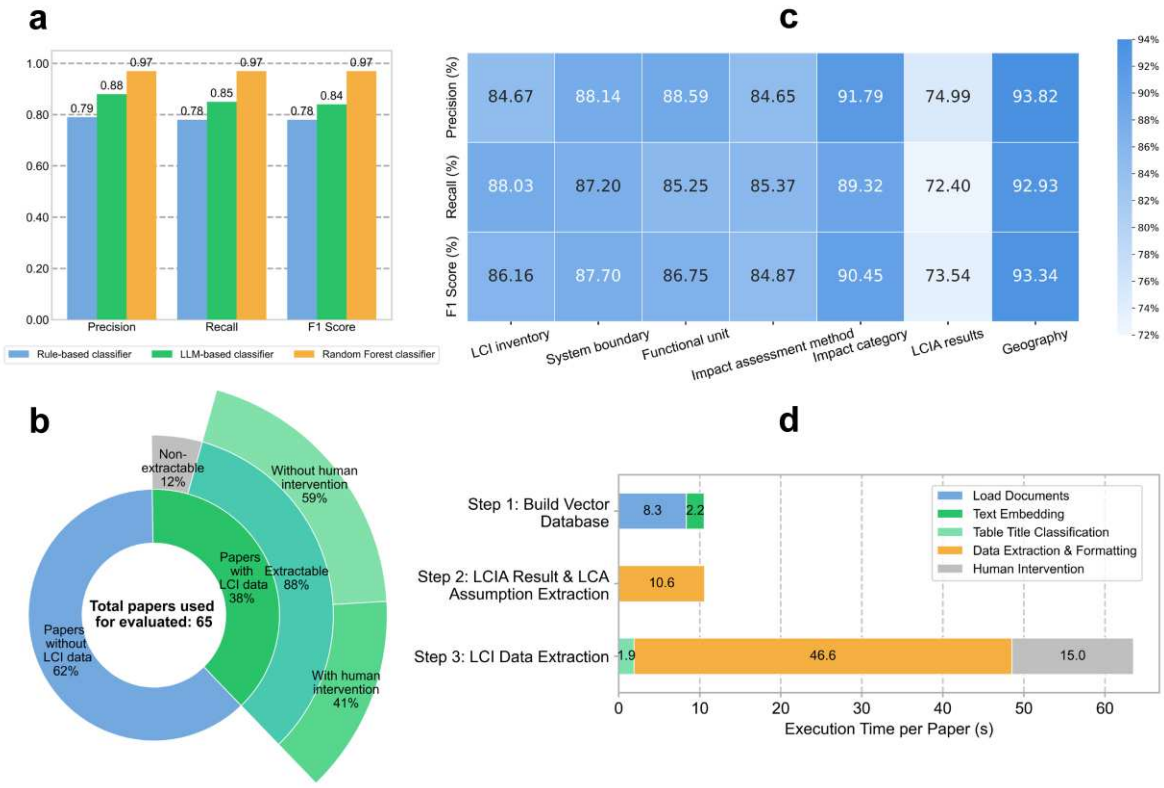


Figure 3. Multifaceted performance analysis of LCA data mining processes. (a) Comparison of classification performance among rule-based, LLM-based, and Random Forest classifiers. (b) Distribution of evaluated papers with and without LCI data, including proportions of extractable and human-assisted extraction cases. (c) Heatmap illustrating precision, recall, and

F1-score percentages for LCA domain data extraction across seven metadata categories. (d) Average execution time per paper of each step, highlighting their relative efficiency.

During the pipeline evaluation, 25 out of 65 papers provided LCI data (Figure 3b). Among them, the LCI tables of three papers could not be extracted, even though the RFC accurately identified their locations. Two of these cases (Fazeni-Fraisl & Lindorfer, 2023; Lopes et al., 2019) resulted from PDF parsing errors, where the parser (i.e., PyMuPDF4LLM in this study) omitted table structures within the PDF files. However, we tested other open-source PDF parsing tools, such as PyPDF and PDFMiner, which successfully extracted the table content as plain text. Although analyzing the detailed reasons for these parsing failures lies beyond the scope of this study, it highlights the importance of addressing such challenges to ensure the reliability of RAG systems (Adhikari & Agarwal, 2025). Moreover, we observed an example in Moretti et al. (2020), where the LCI tables (Table 2 and Table 3) were successfully parsed but returned no extractable content from the LLM. This retrieval failure is potentially caused by the loss of structural information (e.g., number of columns and rows) during the table serialization process, in which two-dimensional tables are converted into a linear, sequential text format (e.g., Markdown). Sui et al. (2024) emphasized that the capability of LLMs to interpret structured tables heavily depends on the quality of the serialized representations, and that errors such as omitting marked partitions or altering the input order can lead to substantial drops in model performance. In addition, Wu et al. (2025) surveyed current table representation strategies and found that the existing serializers struggle to handle certain types of tables, such as those with intricate structures and long, multi-line text, similar to this example, since linearizing such complex layouts often results in incomplete or distorted representations. Therefore, the authors recommended image-based representations as a promising alternative due to their superior ability to preserve structural layouts.

For the remaining studies with extractable LCI content, 41% still required human intervention to identify and provide necessary metadata for autonomous operation, due to errors made by the LLM in detecting this information. We observed that nearly all of these papers presented their LCI tables in a hierarchical format, in which column headers often

denote different product systems or technology scenarios, and row-group headers represent life-cycle stages or sub-processes within the main system, reflecting the way LCI blocks are created in LCA practice (Saavedra-Rubio et al., 2022). Although this hierarchical structure organizes data compactly, it poses considerable challenges for LLM-based table semantic extraction, as models are prone to misinterpreting multi-level structures or omitting crucial headers (Bindini et al., 2025; Wu et al., 2025; Zhang et al., 2024). Table augmentation is one strategy to mitigate these issues by enriching the input table or query with additional knowledge to improve LLM comprehension of complex tables (Su et al., 2024). For example, similar to the method used in this study, Sui et al. (2024) proposed a self-augmented prompting approach in which the LLM first generates a structural description of the table and then incorporates it into the prompt to guide reasoning, yielding better performance than prompting without such structural signals. Recent studies have also shown that adopting reasoning chains, such as chain-of-thought (CoT) and self-consistency prompting, can improve LLM performance in tabular reasoning tasks (Wang et al., 2024; Jin & Lu, 2023). For instance, Wang et al. (2024) used CoT prompting to iteratively generate intermediate tables that preserve the underlying structure cues needed for answering a query, thereby improving reasoning accuracy compared to traditional free-form text reasoning. Another potential challenge arises when LLMs process LCI tables that span multiple pages, such as Table S2-4 in Lin et al. (2023), since LLMs often struggle to interpret multi-page tabular data (Aggarwal et al., 2025). Thus, our pipeline allows users to manually provide the full list of corresponding page numbers in the prompts to guide the model's processing. Developing meta-prompting strategies, as proposed by Suzgun & Kalai (2024), to enable LLMs to analyze document structure (e.g., table layout, page range) is a promising solution for automating this process.

When the required metadata were correctly identified, the model achieved impressive average precision (84.67%), recall (88.03%), and F1-score (86.16%) in LCI data extraction (Figure 3c). The high performance can be attributed to the retrieval mechanism used in this study, which not only provides rich metadata to support LLM to reasoning but also constraints retrieval to pages that contain the relevant LCI tables. However, we observed an

extremely high MAPE of 34,810.97%, primarily due to the LLM misinterpreting floating-point numbers expressed in scientific notation. These values account for approximately 6% of the evaluation dataset, and excluding such cases reduces the MAPE substantially to 26.38%. Another common error observed in the LLM outputs was row misalignment and row omission, mainly caused by the model misidentifying or incompletely detecting row boundaries. Converting LCI tables into images and applying layout embedding can further improve performance, as it provides LLMs with the spatial layout information present in the table images to support reasoning (Lu et al., 2025). Overall, these results indicate that the LLM can handle diverse and complex LCI table formats and extract LCI content with high semantic accuracy; however, its numerical fidelity remains sensitive to decimal number formatting (e.g., scientific notation). Similarly, the model demonstrated strong performance in extracting LCA modeling assumptions, achieving average F1-scores ranging from 84.84% to 93.34%. In contrast, LCIA results extraction yielded a lower average F1-score of 73.54% with a MAPE of 6.06%. This relatively lower F1-score was mainly due to the model failing to identify relevant information (i.e., reported as "not mentioned") or retrieving incorrect information (e.g., extracting comparative statements rather than absolute LCIA results), which could be further improved through prompt optimization. The identified error types and their corresponding potential mitigation strategies are summarized in Table 2.

Figure 3d demonstrates the average execution time per paper for each step in the proposed pipeline. Step 1 (vector database construction) requires approximately 13.7 s per paper, with most of the time consumed by document parsing (8.3 s). Step 2 (LCIA results and LCA assumption extraction) operates in a fully automated manner and requires 10.6 s per paper for data retrieval and formatting. By comparison, Step 3 (LCI data extraction) needs an additional 15 s of human intervention, estimated based on empirically measured average human involvement time per paper. This step is also the most computationally demanding for the LLM due to the need to handle long and complex LCI tables, resulting in a total execution time of 63.5 s per paper. These results underscore the importance of improving the efficiency of Step 3, such as reducing human intervention, to accelerate the overall system.

Table 2: Summary of common errors in LCA data extraction and their potential mitigation strategies.

Tasks	Error types	Description	Potential mitigation strategies
LCI data extraction	PDF parsing failure	Table structures are omitted or not detected by the PDF parser.	Use alternative or ensemble PDF parsers; improve low-level layout detection; adopt image-based table representations.
	Table serialization issues	Loss or distortion of two-dimensional table structure during linearization.	Improve structure-aware serialization; preserve row/column delimiters; use image-based inputs.
	Missing table metadata	Incomplete or missing detection of essential metadata (e.g., column headers and stub headers).	Manual metadata provision; table augmentation; self-augmented prompting; chain-of-though prompting.
	Multi-page LCI tables	LCI tables spanning multiple pages, leading to loss of contextual continuity.	User-specified page ranges; meta-prompting; heuristic-based multi-page table merging.
	Numerical value parsing issues	Misinterpretation of decimal points or scientific notation.	Numeric post-validation; explicit numeric parsing rules; unit-aware prompting.
	Row misalignment and row omission	Misplaced values or missing rows due to incorrect row boundary detection.	Layout-aware reasoning; image-based extraction with layout embeddings.
LCIA results & LCA modeling assumption extraction	Information misidentification	Failure to identify relevant information or retrieval of incorrect content.	Prompt optimization (e.g., stricter output constraints).

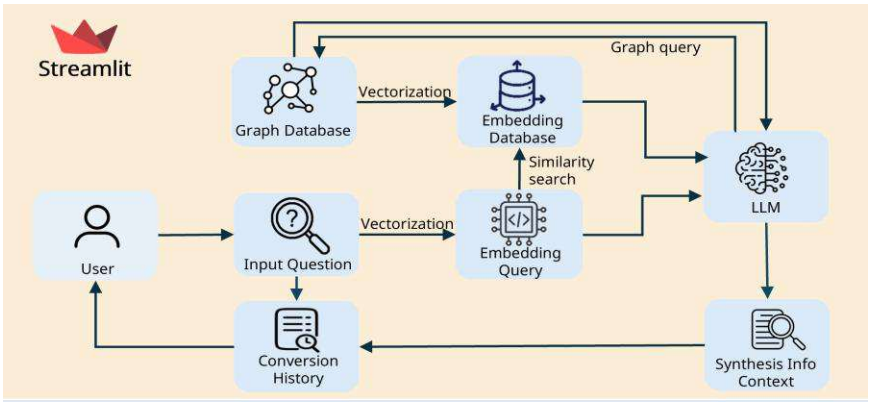
426

3.2 Evaluation of query system performance

When evaluated on a test set of 50 multi-hop questions based on the LCAKG, GraphMAQA achieved a substantially higher F1-score (75.18%) than GraphQA (56.98%), primarily due to more accurate Cypher query generation. GraphQA relies solely on entity identification from user queries to initiate graph traversal, making it more sensitive to mismatch errors (e.g., case sensitivity and formatting inconsistencies). In contrast, GraphMAQA integrates vector-based similarity search with question rewriting to better align user queries with Cypher's strict formalization rules and the instance values stored in the KG, thereby reducing the risk of invalid queries. However, we observed that similarity search can still produce retrieval errors, particularly when queries contain multiple entities. For example, a query requesting LCI data for 'bio-based methanol production' incorrectly retrieves data from the 'natural gas-based methanol production' instead of the more relevant "biomass-to-methanol (BTM) production." This finding is consistent with a recent study by Zhang et al. (2025), which emphasized that embedding multi-entity sentences into a single vector can dilute the semantic representations of individual key entities. Furthermore, Wu et al. (2023) highlighted that a single vector is insufficient for multi-entity queries and suggested the use of multi-vector representations that encode distinct entities as separate vectors for similarity search, although efficiently handling such multi-vector queries remains an open research challenge.

After receiving a user query or a rewritten question, the LLM identifies relevant entities and generates corresponding Cypher queries. We found that the LLM used in this study (i.e., OpenAI's GPT-4o) exhibits strong reasoning capabilities, enabling it to handle multi-hop questions that require complex logical inference. In line with prior work, Mandilara et al. (2025) showed that large-scale models, such as GPT-4o, are better suited than small or medium-sized models for answering multi-hop questions in text-to-Cypher tasks. To further improve the accuracy of Cypher query generation, recent studies have explored approaches such as schema-guided prompting to more effectively represent the KG schema (Wan et al., 2025) and fine-tuning LLMs to reduce hallucinations (Ozsoy et al., 2025; Zou et al., 2024).

Figure 4 illustrates example user interactions with the query system implemented using Streamlit.



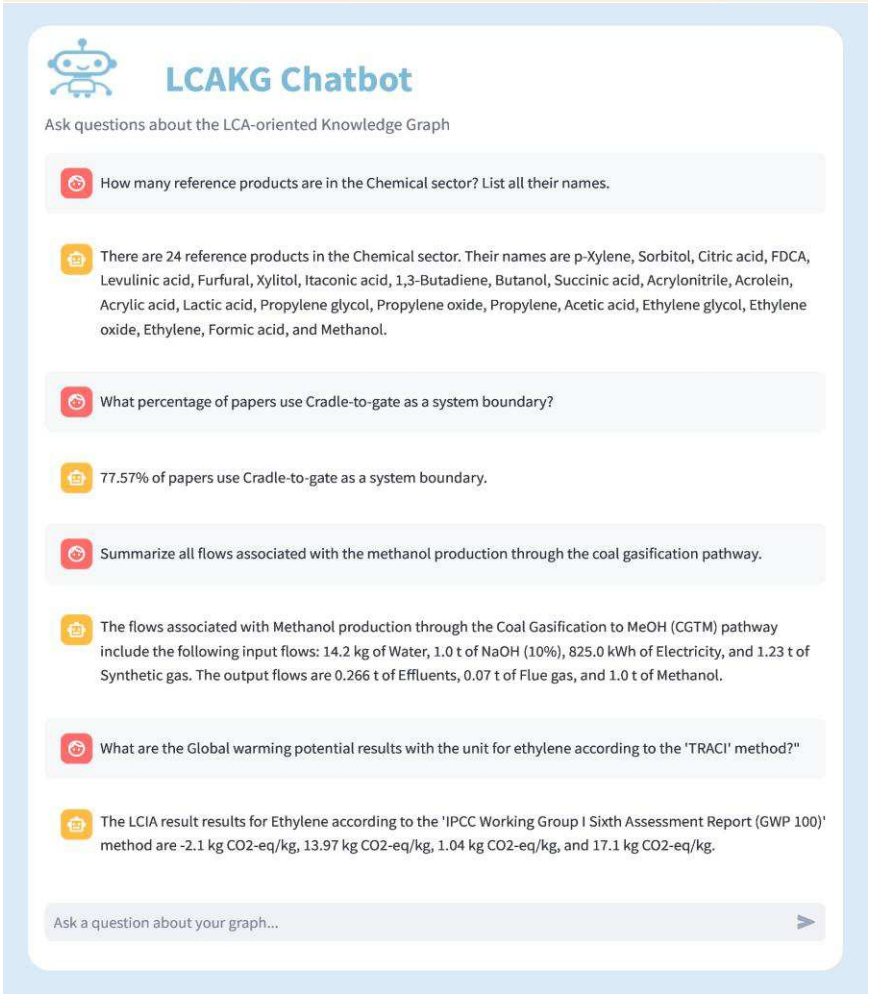


Figure 4. Integrated workflow of the LCAKG chatbot, illustrating the transformation of the graph database into a context-aware dialogue system and example interactions with the data-driven chatbot implemented using Streamlit.

3.3 Case study results

Using the proposed knowledge-retrieval pipeline, we extracted 133 inventory tables and 289 GWP results covering distinct synthesis pathways for 17 chemicals from 107 research studies. To ensure the consistency and reliability of the LCAKG, the extracted data were validated through human review to address the issues identified above (e.g., numerical decimal errors, row misalignment, and row omission). Additionally, we found that certain inherent formatting characteristics of LCI tables can lead to incomplete extraction, requiring extra human effort to fill missing entries. These issues include: (1) missing reference flows (e.g., reference flows only reported in table titles); (2) incomplete inventories (e.g., inputs and outputs reported in separate tables); and (3) non-machine-readable tables (e.g., poorly structured tables). Detailed examples of these cases are provided in Table S4. Following validation, a normalization step was employed to address inherent variability in LCA-domain terminology (e.g., multiple expressions of system boundary). Although this process was performed manually in this study, it could be accelerated by leveraging approaches such as building a domain thesaurus for keyword matching (Peng et al., 2024) and NLP-based entity linking techniques (Schäfer et al., 2024).

We implemented the normalized dataset in Neo4j to construct the LCAKG, which comprises 6,991 nodes and 7,662 relationships. Example query results are shown in Figure 5. Given the inherent structure and interconnected nature of graph databases, the KG enables a wide range of analytical capabilities, including direct querying, causal inference, and graph-based analytical techniques (Greif et al., 2025). Leveraging the relationships encoded in the LCAKG, we further analyzed data extracted from the literature (e.g., GWP results for selected chemicals), as detailed in the SI.

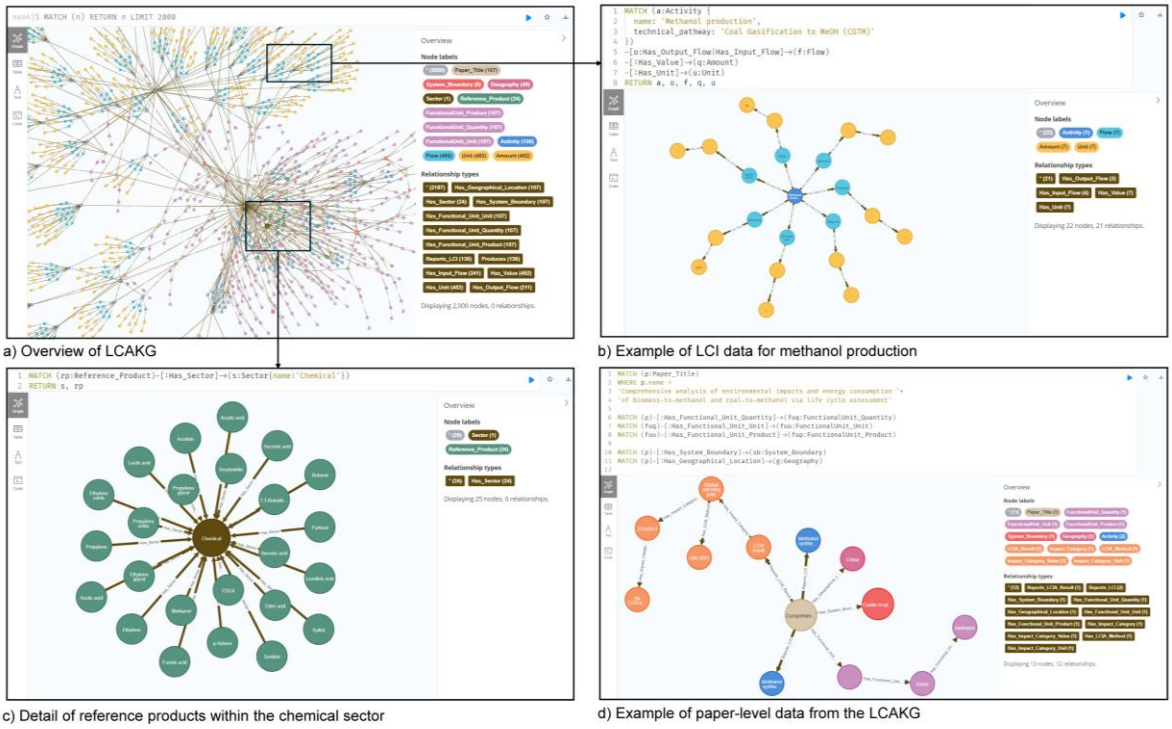


Figure 5. Visualization of the LCAKG in Neo4j.

3.4 Limitations and future work

Despite the promising results of the proposed framework, several limitations should be acknowledged to contextualize the findings. First, the extraction of LCI tables still requires partial human intervention for identifying missing metadata (e.g., multi-level headers) during extraction and for correcting numerical or structural inconsistencies during validation. This reliance on manual review limits the full automation potential of the pipeline. Second, compared with established background databases (e.g., Ecoinvent) that provide comprehensive product and activity information, the proposed pipeline (particularly the prompt design) does not fully capture detailed descriptive metadata (e.g., product and process descriptions) reported in the source literature. The absence of such metadata may reduce the usability of the LCAKG for users who require well-specified and contextualized LCA data. Third, the current pipeline focuses on extracting information from textual and tabular content, while ignoring figure-based data that may contain valuable information (e.g., LCIA results).

Finally, the framework was evaluated exclusively on LCA studies of chemical production, and its transferability to other application domains (e.g., construction, agriculture, or the energy sector) remains to be evaluated.

To improve the robustness and applicability of the proposed framework, several emerging techniques (e.g., image-based inputs and chain-of-though prompting) can be implemented, as summarized in Table 2. Beyond these methodological enhancements, we identify three important directions for future work. First, our study shows that LCI data reported in the literature are rich and informative. However, most LCI data are still presented in human-oriented table layouts that lack consistent schemas and standardized metadata fields, which hinders efficient data extraction. Although existing studies have proposed detailed guidelines for LCI data collection (Saavedra-Rubio et al., 2022), there is less guidance on how LCI tables should be presented in academic publications. Therefore, future research could focus on developing a generic, machine-readable, and community-accepted LCI representation framework to facilitate LCI data retrieval from the literature (Malek et al., 2024). Second, rather than executing the sub-processes (e.g., data extraction, validation, KG construction, and interaction) sequentially as in this study, future work could further exploit LLM capabilities with modular, multi-agent processes to support end-to-end KG management. By deploying LLM-based agents that each focus on a specific subtask, the multi-agent system can collaboratively perform planning, decision-making, and action execution in an autonomous and coordinated manner (Guo et al., 2024). This approach could reduce reliance on human intervention and improve robustness in handling heterogeneous and complex LCA data extraction and management tasks. Third, the constructed LCAKG opens opportunities for advanced graph-based analysis, particularly through graph neural networks (GNNs). GNNs are deep learning-based methods that operate directly on graph-structured data by propagating and aggregating information from neighboring nodes (Ye et al., 2022). By supporting tasks such as link prediction and node classification, GNNs can enable the inference of missing inventory data and the estimation of environmental impacts within the LCAKG, thereby helping to mitigate long-standing data challenges in the LCA community.

4. Conclusion

This study presents a robust and scalable framework that integrates the LLM with an ontology-driven KG to enable efficient extraction and management of LCA data from scientific literature. By combining advanced RAG architectures with tailored prompt engineering, the workflow addresses inherent challenges arising from heterogeneous document formats and complex layouts, enabling the extraction of diverse LCA data types (e.g., LCI data, LCIA results, and LCA modeling assumptions) from PDF files with high precision, recall, and F1-scores. The extracted information is formalized through an LCA-oriented ontology and instantiated in a graph-based LCAKG, which supports structured querying and advanced graph-based reasoning. Additionally, we developed a data retrieval system that integrates similarity search with LLM-based text-to-Cypher reasoning, allowing users to retrieve information from the LCAKG through natural language queries. The framework was demonstrated using LCA studies on chemical production but is generalizable across other application fields.

To reduce barriers to adoption, the framework was integrated into a Streamlit-based user interface that enables users with minimal technical expertise to operate the system. The customizable prompts and a flexible system design further position the framework as a valuable tool for broader scientific applications, such as data collection and synthesis in literature reviews. Future work could focus on improving extraction robustness by adopting advanced prompting strategies (e.g., chain-of-thought prompting) and leveraging multimodal capabilities for image-based data extraction. Besides, developing LLM-based multi-agent systems represents a promising direction for fully automating end-to-end knowledge graph management. Overall, the proposed framework facilitates the development of interoperable, machine-readable LCA databases that support large-scale literature synthesis, data reuse, and advanced computational analysis. Its continued development and open accessibility can foster innovation and collaboration in data-driven sustainability research and beyond.

Data Availability Statement

The ground-truth data and LLM-extracted results are provided in the Supplementary Data S1. The normalized dataset mapped into Neo4j is provided in the Supplementary Data S2. The dataset used for LCI table title classification is provided in the Supplementary Data S3. The pipeline code developed in this study is available in the following GitHub repository: https://github.com/Zirui0102/LCAKG_RAG.

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